

Peer-adult network structure and suicide attempts in 38 high schools: implications for network-informed suicide prevention (10,291 students)

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To date, suicide prevention has focused primarily on individual-level psychiatric risk factors and on strategies to refer and treat high-risk youth. Yet researchers and policymakers recommend strengthening positive social bonds because social integration can reduce suicidal behavior in youth

To our knowledge, this is the first study of any adolescent health problem, including suicidal behavior, that integrates adult connections into friendship networks at a school population level.

Network interventions addressing these processes, such as maximizing youth–adult connections schoolwide and heightening influence of youth with healthy coping, could create more protective schools. The present study is the first empirical evidence linking more integrated youth–adult networks (i.e., students sharing trusted adults with friends) to lower Suicide Attempts.

Youth with positive bonds to family, peers, and/or school are generally less likely to consider or attempt suicide

Specifically, schoolwide rates of Suicide Ideation were higher in schools where there were more peer isolates, fewer friendship ties, and smaller friendship groups that were less dense, and when students' friends were less likely to be friends with each other.

Schools with friendship networks reflecting greater integration and cohesion had lower rates of Suicide Ideation and Suicide Attempts. Students with more friendship ties and who were part of larger, interconnected friendship groups were less likely to report SI and SA;

Social networks provide the mechanisms for the diffusion of norms and practices and the context for peer group monitoring and support. Network characteristics influence the spread of many health behaviors. By focusing on patterns of relationships, network methods can clarify the contexts in which relationships form and exert influence on others.

Cohesive youth–adult networks may promote more help-seeking for students and for their friends with whom they share a common bond to a supportive adult.

Maximizing protective bonds across school populations, increasing opportunities for group cohesion including integrated youth–adult networks, and promoting influence of youth with healthy coping. Our findings suggest effective interventions will involve youth and adults.

Fruitful areas for future work include identifying new strategies for integrating adolescents' peer and adult networks and increasing accessibility of competent adults.

Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial

Peter A. Wyman, PhD; Anthony R. Pisani, PhD; C. Hendricks Brown, et al.

The Wingman-Connect program used group skill building for cohesion, shared purpose, and managing career and personal stressors. Wingman-Connect is the first universal prevention program to reduce suicidal ideation and depression symptoms in a general Air Force population. Group training that builds cohesive, healthy military units is promising for upstream suicide prevention.

The study's findings validate the underlying network health model: stronger bonds within a more cohesive healthy class reduced suicidal ideation and depression symptoms. These findings suggest that Wingman-Connect classes became increasingly unified around healthy norms and encouraged classmates who were vulnerable to mental health or occupational problems at a key juncture of military training, in addition to meeting their needs for belonging. Airmen learn skills to grow and sustain protective factors essential to job success, mental health and reduced suicide risk (Kinship, Purpose, Guidance, Balance). Class exercises create more cohesive units with skills extended into group culture. Wingman Connect training is delivered to organizational units using interactive exercises to build key group and individual protective factors (Four Cores):

- Group cohesion and belonging (Kinship)
- Meaning and value in work, life and role (Purpose)
- Informal and formal support and help-seeking (Guidance)
- Diversity of activities that give strength and perspective in times of transition and difficulty (Balance)

Active Learning and Transfer of Training to Life

- Drawing out 'real-world' strengths from participants as a primary method to teach Kinship, Purpose, Guidance, & Balance.
- High energy activities maintain interest, motivation, and a shared positive emotion.
- 'Real-world' exercises inside and outside of training reinforce the application of skills in groups and in individual activities of daily life.

Research Outcomes – Evidence Based Key Findings include:

- Increased class cohesion, morale, positive bonds, and acceptability of help seeking
- Reduction in suicide risks scores, depression, and reactivity to anger
- Nearly 50% reduction in occupational impairment (corrective training, negative counseling)

Wingman-Connect Program increases social integration for Air Force personnel at elevated suicide risk: Social network analysis of a cluster RCT

Peter A. Wyman, Trevor A. Pickering, Anthony R. Pisani, Ian Cero, Bryan Yates, et al.

Wingman-Connect Program built enhanced suicide protection into unit relationship networks and counteracted standard drift towards disconnection for at-risk Airmen, despite no explicit content targeting connections specifically to at-risk Airmen. Findings support a growing case for the unique contribution of group-level interventions to improve social health of broader populations.

The present study thus tested whether Wingman-Connect Program achieved the objective of strengthening the network structure of unit relationships over time, including socially integrating those currently at elevated risk for suicide. This is an especially important hypothesis since W-CP training has no content encouraging Airmen to form connections with at-risk peers. The Wingman-Connect Program trains all members of natural organizational units together to strengthen group bonds and the diffusion of suicide-protective norms and practices. Skill-building activities focus on relationships and practices supportive of mental health. Specific modules and activities build healthy relationships and accountability (Kinship), meaning and value in work and life (Purpose), informal and formal help-seeking (Guidance), and activities that give strength (Balance). Group skill-building activities identify strengths of all group members, and members learn how a strong network supports all members' health and fitness. W-CP uses high energy activities and peer-to-peer teaching; exercises inside and outside of training promote adoption of skills into unit culture.

Airmen at elevated suicide risk vs. not at risk were significantly more likely to have elevated depression symptoms (93.9% vs. 12.2%), loneliness, emotion regulation difficulties, anger, and scores on military functional impairment-social scale.

Wingman-Connect counteracted the typical **drift** towards disconnection for Airmen at elevated suicide risk. Six-month findings again showed that W-CP offset the typical trajectory of decreasing connections for Airmen at elevated suicide risk.

These findings are, to our knowledge, the first to show that a suicide prevention program for small personnel units improved significantly the group relationship network and socially integrated members over time.

This study extends knowledge of the social and relationship impact of the Wingman-Connect Program as it decreased suicide risk for Airmen in training.

The current attention on the detection and treatment of suicidal military members is necessary, but ultimately insufficient to meet the needs of many suicidal service members. Results from our study suggest that the status quo progression for many service members at elevated suicide risk is one of worsening disconnection, evidenced by the trajectory of declining valued connections found in the active control condition. One in 10 Airmen at elevated suicide risk in the control condition in our study (10%) had no connections to other members at the one-month follow-up, approximately two months after technical training classes began. Wingman-Connect Program counteracted the expected drift towards disconnection for Airmen at elevated suicide risk by targeting the unit's relationship network itself. These groups built

enhanced suicide protection into their relationship networks, with the most consistent benefits for Airmen at elevated risk of suicide and for those starting with fewest connections. In more connected, healthier units, vulnerable members can borrow strength from adaptive members. Stronger units also promote job fitness, as shown by W-CP's impact reducing occupational problems, an effect that may reduce future problems that precipitate new instances of suicidal thoughts and behaviors. In that sense, W-CP may specifically assist vulnerable Airmen in overcoming barriers to forming meaningful social connections. Wingman-Connect Program's interactive training created more dynamic relationship networks. Airmen in W-CP groups made more changes in who they named as valued connections, even as the overall average number of valued connections increased. The largest dynamic changes were for Airmen with elevated suicide risk in W-CP groups, who made on average three times more new valued connection nominations.

An Outcome Evaluation of Suicide Prevention Program Delivered by Adolescent Peer Leaders in High Schools Am J Public Health. 2010 September; 100(9): 1653–1661

[Peter A. Wyman](#), PhD, [C. Hendricks Brown](#), PhD, et al.

Training improved the peer leaders' adaptive norms regarding suicide, their connectedness to adults, and their school engagement, with the largest gains for those entering with the least adaptive norms. Trained peer leaders in larger schools were 4 times as likely as were untrained peer leaders to refer a suicidal friend to an adult. Among students, the intervention increased perceptions of adult support for suicidal youths and the acceptability of seeking help.

The norms most strongly enhanced through the intervention were students' perceptions that adults in their school can provide help to suicidal students and the acceptability of seeking help from adults.

Adolescents are far more likely than are adults to be aware of suicidal behavior in their friends; increasing students' partnering with adults to help suicidal peers may be a key process for reducing adolescent suicidal behavior.

By contrast, a recent study of gatekeeper training for adult staff in secondary schools found minimal change in adult-student communication about suicide

This is quite similar to recent evidence that gatekeeper training enhances school staff members' ability to talk to students about suicide only for adults already addressing issues of suicide with students.